PROJECT REVIEW 2

Multimodal Hallucination Firewall

Evidence-Based AI Answer Verification

Progress report: interactive web prototype, evidence analysis and current limitations

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SUBJECT	Machine Learning
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The prototype has progressed from the proposed verification framework in Review 1 to a working web interface that accepts visual evidence and tests whether an AI-generated answer is consistent with it.

1. Progress since Project Review 1

The first review described the intended direction: combine visual and textual evidence with machine-learning-based risk scoring to identify answers that are unsupported or contradictory. The current milestone is an interactive Streamlit website that demonstrates the verification stage using an uploaded image, a question and a supplied AI answer.

What is implemented in the current prototype

COMPONENT	CURRENT ROLE
Image and prompt input	The user uploads an image and enters a question to define what should be checked.
Answer verification input	The user enters an AI-generated answer; the prototype evaluates its consistency with the image.
CLIP-based visual evidence	The interface displays image–text/object evidence scores, including present-versus-absent evidence for a queried object.
SVM risk classification	A support/risk estimate is shown as a hallucination probability, together with a firewall verdict.
Interpretable result page	The verdict screen presents the queried object, evidence scores, risk level and a short explanation of the decision.

Current verification flow

1. Upload image	2. Enter question	3. Enter AI answer
Visual input	Defines the claim to inspect	Candidate response
4. Extract CLIP features	**5. SVM risk assessment**	**6. Firewall verdict**
Image–text/object signals	Estimates unsupported-answer risk	Supported or hallucinated

Scope note: the screenshots demonstrate the verification interface and its decision pipeline. They do not establish that document retrieval, RAG-based evidence retrieval, or automatic answer generation is already integrated into the website.

2. Demonstration A — Contradictory answer

Question: “Is there a phone in the image?”

The uploaded scene contains a laptop on a table and a small phone near the bottom-centre area, below the laptop. In this run, the supplied answer says that the phone is not visible. The firewall marks the answer as **HALLUCINATED** because the answer conflicts with the object-level evidence shown by the prototype.

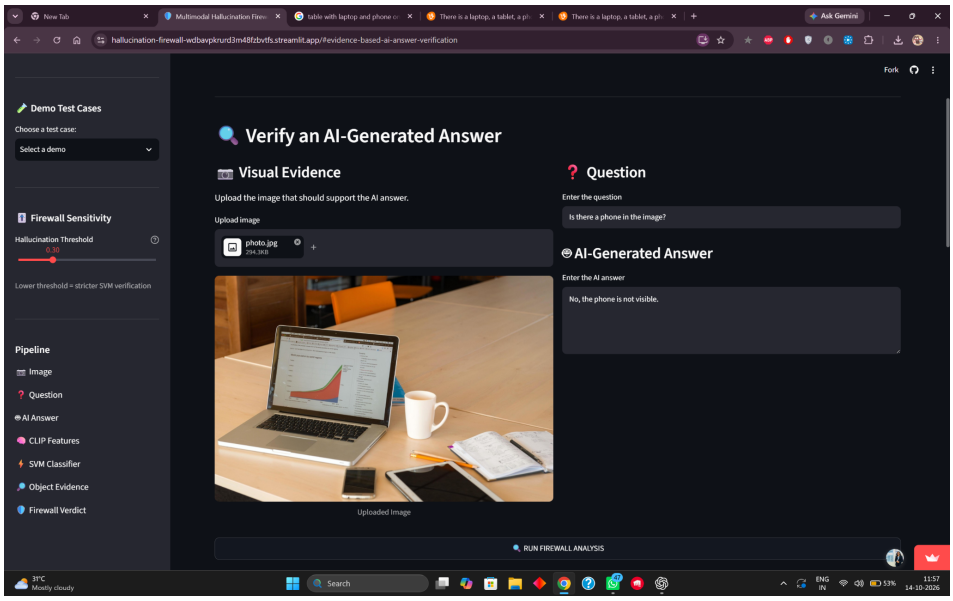


Figure 1. Input screen: uploaded image, question and answer denying the phone’s presence.

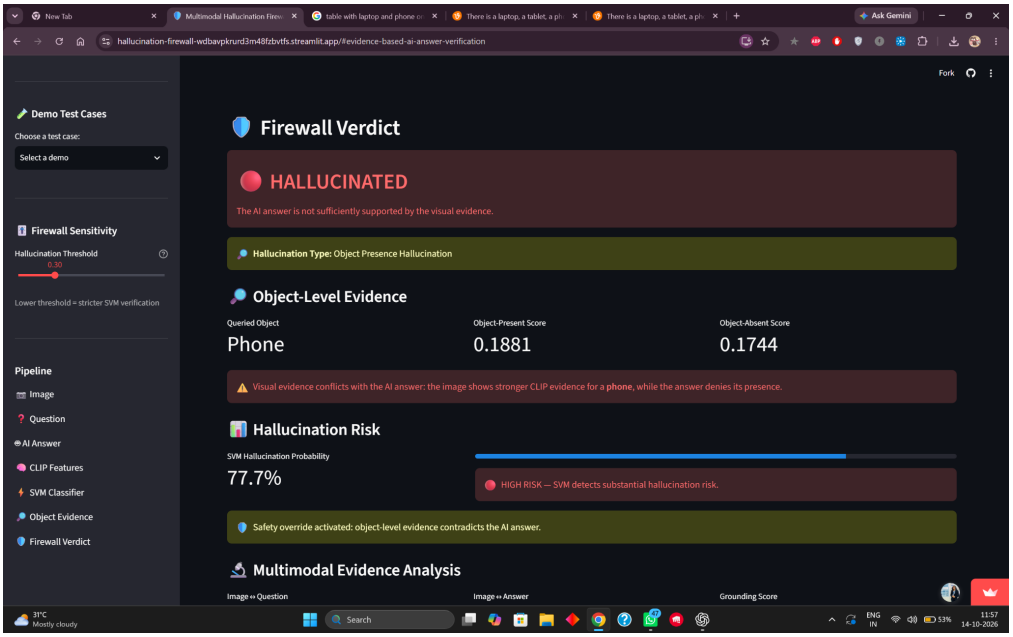


Figure 2. Verdict screen: the prototype flags an object-presence contradiction.

3. Demonstration B — Visually supported answer

The same image and question are used, but the answer now states that the phone is visible near the bottom centre, below the laptop. The firewall labels this response **SUPPORTED**. Using the same scene for both runs makes the comparison useful: the answer changes while the visual evidence remains fixed.

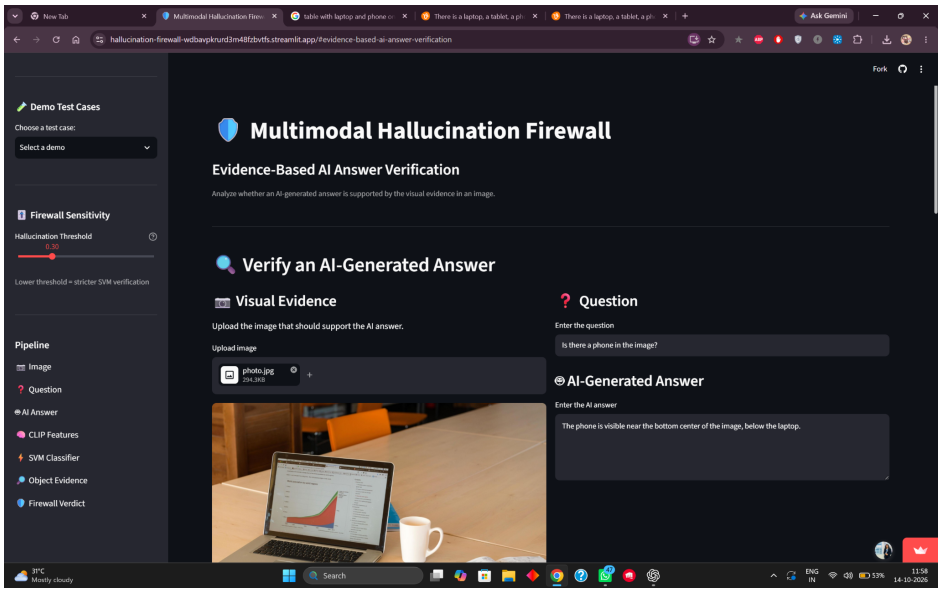


Figure 3. Input screen: answer identifies the phone's location in the image.

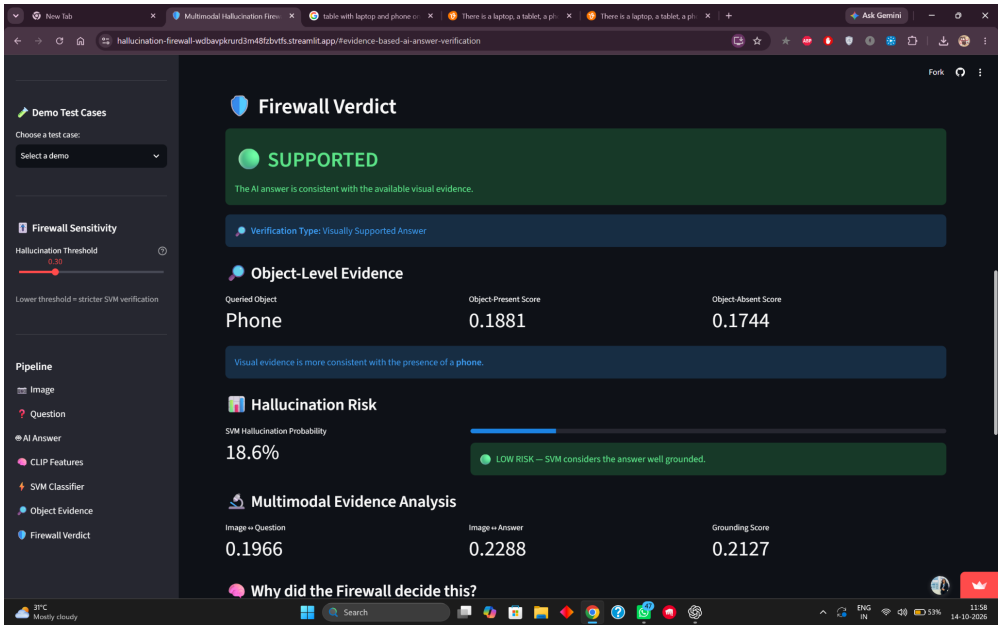


Figure 4. Verdict screen: the prototype accepts the answer as visually supported.

4. Results from the two prototype runs

The following values are read directly from the supplied screenshots. They describe these two demonstrations only and should not be interpreted as benchmark performance.

MEASURE	Contradictory answer	Supported answer
Firewall verdict	HALLUCINATED	SUPPORTED
Queried object	Phone	Phone
Object-present score	0.1881	0.1881
Object-absent score	0.1744	0.1744
SVM hallucination probability	77.7%	18.6%
Additional displayed values	Object-presence conflict	Image–question: 0.1966 Image–answer: 0.2288 Grounding: 0.2127

Interpretation

The prototype produces different risk outputs for the two answer variants: 77.7% for the denial and 18.6% for the location-specific answer. The present score (0.1881) is only slightly higher than the absent score (0.1744), a margin of 0.0137. This is a narrow separation, so the object-level result should be treated as a useful signal rather than definitive proof. The screenshots show that the interface can expose evidence and a verdict; they do not show whether these scores are calibrated or how often the classifier is correct across a dataset.

Important distinction: CLIP similarity values are not probabilities of truth. The displayed SVM risk is a model output and needs validation and, ideally, calibration before being presented as a reliable confidence estimate.

5. Current limitations and next improvements

The next phase should focus on reliability and measurable evaluation, not only adding more interface elements.

1. Expand the labelled test set

Build a diverse collection of image–question–answer examples covering multiple object categories, absent objects, occlusion, small objects, cluttered scenes, and hard negatives. Record labels consistently.

2. Train and validate the SVM properly

Use a stratified train/validation/test split, prevent near-duplicate leakage, and apply cross-validation on the training portion. Compare against a simple baseline and check class balance.

3. Report real evaluation metrics

Measure precision, recall, F1-score, ROC-AUC and a confusion matrix on held-out data. Include threshold-sensitivity results and explain which error matters most for the intended use.

4. Calibrate the risk score

Check whether predicted risk corresponds to observed error frequency. Tune the decision threshold using validation data, and consider an “uncertain / needs review” outcome when evidence is weak.

5. Strengthen evidence grounding

Move beyond a single object-level similarity comparison where possible: extract answer claims, check each claim against visual evidence, and show which claim triggered the warning. Add explicit handling for ambiguous or unrecognisable content.

6. Improve robustness and usability

Test different image sizes, file types and upload failures; add clear error messages, reproducible logs and a short explanation of the evidence behind each verdict.

Planned evaluation table

Evaluation item	What it will establish
Held-out test set	Performance on examples not used for fitting or threshold selection.
Confusion matrix + precision/recall/F1	Types of errors and the balance between missed hallucinations and false alarms.
Threshold comparison	How the verdict changes as the risk threshold is adjusted.
Ablation study	Whether CLIP/object evidence and SVM scoring each contribute useful signal.

6. Summary

Project Review 2 demonstrates a working web interface for evidence-based answer verification. In the supplied example, the firewall distinguishes an answer that denies a visible phone from an answer that correctly identifies it, and presents the decision with object-level scores and an SVM risk estimate. The immediate priority is to establish reliability through a larger labelled dataset, disciplined validation, calibrated thresholds and transparent metrics before making broader claims about hallucination detection.